

Лабораторная работа №4

Первоначальное конфигурирование сети

Поляков Г. С.

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Российский университет дружбы народов, Москва, Россия

Информация

- Поляков Глеб Сергеевич
- студент
- Российский университет дружбы народов

Выполнение работы

Размещение устройств согласно схеме

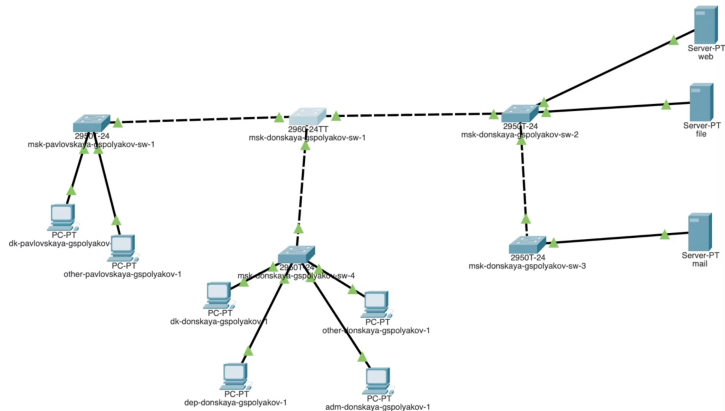


Рис. 1: Схема сети

Конфигурация msk-donskaya-gspolyakov-sw-1 1

```
Switch(config)#hostname msk-donskaya-gspolyakov-sw-1
msk-donskaya-gspolyakov-sw-1(config)#
msk-donskaya-gspolyakov-sw-1(config)#
msk-donskaya-gspolyakov-sw-1(config)#int
msk-donskaya-gspolyakov-sw-1(config)#interface vl
msk-donskaya-gspolyakov-sw-1(config)#interface vlan2
msk-donskaya-gspolyakov-sw-1(config-if)#no sh
msk-donskaya-gspolyakov-sw-1(config-if)#no shutdown
msk-donskaya-gspolyakov-sw-1(config-if)#ip add
msk-donskaya-gspolyakov-sw-1(config-if)#ip address 10.128.1.2 255.255.255.0
msk-donskaya-gspolyakov-sw-1(config-if)#exit
msk-donskaya-gspolyakov-sw-1(config)#ip def
msk-donskaya-gspolyakov-sw-1(config)#ip default-gateway 10.128.1.1
msk-donskaya-gspolyakov-sw-1(config)#line vty 0 4
msk-donskaya-gspolyakov-sw-1(config-line)#pass
msk-donskaya-gspolyakov-sw-1(config-line)#password cisco
msk-donskaya-gspolyakov-sw-1(config-line)#login
msk-donskaya-gspolyakov-sw-1(config-line)#exit
msk-donskaya-gspolyakov-sw-1(config)#console 0 password cisco
^
% Invalid input detected at '^' marker.

msk-donskaya-gspolyakov-sw-1(config)#console 0 password cisco
^
% Invalid input detected at '^' marker.

msk-donskaya-gspolyakov-sw-1(config)#console 0
^
% Invalid input detected at '^' marker.

msk-donskaya-gspolyakov-sw-1(config)#line console 0
msk-donskaya-gspolyakov-sw-1(config-line)#password cisco
msk-donskaya-gspolyakov-sw-1(config-line)#login
msk-donskaya-gspolyakov-sw-1(config-line)#exit
msk-donskaya-gspolyakov-sw-1(config)#enable se
msk-donskaya-gspolyakov-sw-1(config)#enable secret ci
msk-donskaya-gspolyakov-sw-1(config)#enable secret cisco
msk-donskaya-gspolyakov-sw-1(config)#ser
msk-donskaya-gspolyakov-sw-1(config)#service pa
msk-donskaya-gspolyakov-sw-1(config)#service password-encryption
msk-donskaya-gspolyakov-sw-1(config)#username admin privla
msk-donskaya-gspolyakov-sw-1(config)#username admin priv
msk-donskaya-gspolyakov-sw-1(config)#username admin privilege 1 secret cisco
msk-donskaya-gspolyakov-sw-1(config)#ip domain n
msk-donskaya-gspolyakov-sw-1(config)#ip domain-
msk-donskaya-gspolyakov-sw-1(config)#ip domain-n
msk-donskaya-gspolyakov-sw-1(config)#ip domain-name don
msk-donskaya-gspolyakov-sw-1(config)#ip domain-name donskey.ru
msk-donskaya-gspolyakov-sw-1(config)#ip domain-name donskey.ru
```

Конфигурация msk-donskaya-gspolyakov-sw-1 2

```
msk-donskaya-gspolyakov-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-gspolyakov-sw-1.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
    a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-gspolyakov-sw-1(config)#line vty 0 4
*Mar 1 0:12:21.590: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:12:21.590: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-gspolyakov-sw-1(config-line)#trans
msk-donskaya-gspolyakov-sw-1(config-line)#transport in
msk-donskaya-gspolyakov-sw-1(config-line)#transport input ssh
msk-donskaya-gspolyakov-sw-1(config-line)#exit
msk-donskaya-gspolyakov-sw-1(config)#exi
msk-donskaya-gspolyakov-sw-1(config)#exit
msk-donskaya-gspolyakov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-gspolyakov-sw-1#write memory
Building configuration...
[OK]
msk-donskaya-gspolyakov-sw-1#
```

Рис. 3: Конфигурация msk-donskaya-gspolyakov-sw-1 2

Конфигурация msk-donskaya-gspolyakov-sw-2 1

```
Switch(config)#hostname msk-donskaya-gspolyakov-sw-1
msk-donskaya-gspolyakov-sw-1(config)#
msk-donskaya-gspolyakov-sw-1(config)#hostname msk-donskaya-gspolyakov-sw-2
msk-donskaya-gspolyakov-sw-2(config)#int
msk-donskaya-gspolyakov-sw-2(config)#interface vlan2
msk-donskaya-gspolyakov-sw-2(config-if)#no s
msk-donskaya-gspolyakov-sw-2(config-if)#no sh
msk-donskaya-gspolyakov-sw-2(config-if)#no shutdown
msk-donskaya-gspolyakov-sw-2(config-if)#ip ad
msk-donskaya-gspolyakov-sw-2(config-if)#ip address 10.128.1.3 255.255.255.0
msk-donskaya-gspolyakov-sw-2(config-if)#exit
msk-donskaya-gspolyakov-sw-2(config)#ip defau
msk-donskaya-gspolyakov-sw-2(config)#ip default-gateway 10.128.1.1
msk-donskaya-gspolyakov-sw-2(config)#line vty 0 4
msk-donskaya-gspolyakov-sw-2(config-line)#password cisco
msk-donskaya-gspolyakov-sw-2(config-line)#login
msk-donskaya-gspolyakov-sw-2(config-line)#exit
msk-donskaya-gspolyakov-sw-2(config)#line console 0
msk-donskaya-gspolyakov-sw-2(config-line)#pas
msk-donskaya-gspolyakov-sw-2(config-line)#password cisco
msk-donskaya-gspolyakov-sw-2(config-line)#login
msk-donskaya-gspolyakov-sw-2(config-line)#exit
msk-donskaya-gspolyakov-sw-2(config)#enable sec
msk-donskaya-gspolyakov-sw-2(config)#enable secret cisco login?
LINE
msk-donskaya-gspolyakov-sw-2(config)#enable secret cisco
msk-donskaya-gspolyakov-sw-2(config)#ser
msk-donskaya-gspolyakov-sw-2(config)#service pa
msk-donskaya-gspolyakov-sw-2(config)#service password-encryption
msk-donskaya-gspolyakov-sw-2(config)#us
msk-donskaya-gspolyakov-sw-2(config)#username admin pr
msk-donskaya-gspolyakov-sw-2(config)#username admin privilege 1 secret cisco
msk-donskaya-gspolyakov-sw-2(config)#ip domain-name donsкаya.rudn.edu
msk-donskaya-gspolyakov-sw-2(config)#cryp
msk-donskaya-gspolyakov-sw-2(config)#crypto k
msk-donskaya-gspolyakov-sw-2(config)#crypto key g
msk-donskaya-gspolyakov-sw-2(config)#crypt k generate
```


Конфигурация msk-donskaya-gspolyakov-sw-2 2

```
msk-donskaya-gspolyakov-sw-2(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-gspolyakov-sw-2.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
    a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-gspolyakov-sw-2(config)#line vty 4 0
*Mar 1 0:16:55.994: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:16:55.994: %SSH-5-ENABLED: SSH 1.5 has been enabled
                                     ^
% Invalid input detected at '^' marker.

msk-donskaya-gspolyakov-sw-2(config)#tr
msk-donskaya-gspolyakov-sw-2(config)#tra
msk-donskaya-gspolyakov-sw-2(config)#line vty 0 4
msk-donskaya-gspolyakov-sw-2(config-line)#tr
msk-donskaya-gspolyakov-sw-2(config-line)#transport i
msk-donskaya-gspolyakov-sw-2(config-line)#transport input s
msk-donskaya-gspolyakov-sw-2(config-line)#transport input ssh
msk-donskaya-gspolyakov-sw-2(config-line)#exit
msk-donskaya-gspolyakov-sw-2(config)#exit
msk-donskaya-gspolyakov-sw-2#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-gspolyakov-sw-2#writ
msk-donskaya-gspolyakov-sw-2#write mem
msk-donskaya-gspolyakov-sw-2#write memory
Building configuration...
[OK]
msk-donskaya-gspolyakov-sw-2#
```

Конфигурация msk-donskaya-gspolyakov-sw-3 1

```
msk-donskaya-gspolyakov-sw-3(config)#interface vlan2
msk-donskaya-gspolyakov-sw-3(config-if)#no sh
msk-donskaya-gspolyakov-sw-3(config-if)#no shutdown
msk-donskaya-gspolyakov-sw-3(config-if)#ip add
msk-donskaya-gspolyakov-sw-3(config-if)#ip address 10.128.1.4
% Incomplete command.
msk-donskaya-gspolyakov-sw-3(config-if)#ip address 10.128.1.4 255.255.255.0
msk-donskaya-gspolyakov-sw-3(config-if)#exit
msk-donskaya-gspolyakov-sw-3(config)#ip con
msk-donskaya-gspolyakov-sw-3(config)#ip
msk-donskaya-gspolyakov-sw-3(config)#ip d
msk-donskaya-gspolyakov-sw-3(config)#ip de
msk-donskaya-gspolyakov-sw-3(config)#ip default-gateway 10.128.1.1
msk-donskaya-gspolyakov-sw-3(config)#line vty 0 4
msk-donskaya-gspolyakov-sw-3(config-line)#password cisco
msk-donskaya-gspolyakov-sw-3(config-line)#login
msk-donskaya-gspolyakov-sw-3(config-line)#exit
msk-donskaya-gspolyakov-sw-3(config)#line console 0
msk-donskaya-gspolyakov-sw-3(config-line)#passw
msk-donskaya-gspolyakov-sw-3(config-line)#password cisco
msk-donskaya-gspolyakov-sw-3(config-line)#login
msk-donskaya-gspolyakov-sw-3(config-line)#exit
msk-donskaya-gspolyakov-sw-3(config)#ena
msk-donskaya-gspolyakov-sw-3(config)#enable secret cisco
msk-donskaya-gspolyakov-sw-3(config)#se
msk-donskaya-gspolyakov-sw-3(config)#service pa
msk-donskaya-gspolyakov-sw-3(config)#service password-encryption
msk-donskaya-gspolyakov-sw-3(config)#user
msk-donskaya-gspolyakov-sw-3(config)#username admin pr
msk-donskaya-gspolyakov-sw-3(config)#username admin privilege 1 secret cisco
msk-donskaya-gspolyakov-sw-3(config)#ip domain donsкаya.rudn.edu
^
% Invalid input detected at '^' marker.

msk-donskaya-gspolyakov-sw-3(config)#ip domain-name donsкаya.rudn.edu
msk-donskaya-gspolyakov-sw-3(config)#cry
msk-donskaya-gspolyakov-sw-3(config)#crypto k
msk-donskaya-gspolyakov-sw-3(config)#crypto key g
msk-donskaya-gspolyakov-sw-3(config)#crypto key generate r
msk-donskaya-gspolyakov-sw-3(config)#crypto key generate rsa
The name for the keys will be: msk-donskaya-gspolyakov-sw-3.donsкаya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]
```

Конфигурация msk-donskaya-gspolyakov-sw-3 2

```
msk-donskaya-gspolyakov-sw-3(config)#line vty 0 4
*Mar 1 0:23:8.814: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:23:8.814: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-gspolyakov-sw-3(config-line)#tran
msk-donskaya-gspolyakov-sw-3(config-line)#transport in
msk-donskaya-gspolyakov-sw-3(config-line)#transport input s
msk-donskaya-gspolyakov-sw-3(config-line)#transport input ssh
msk-donskaya-gspolyakov-sw-3(config-line)#exit
msk-donskaya-gspolyakov-sw-3(config)#exit
msk-donskaya-gspolyakov-sw-3#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-gspolyakov-sw-3#write me
msk-donskaya-gspolyakov-sw-3#write memory
Building configuration...
[OK]
msk-donskaya-gspolyakov-sw-3#
```

Рис. 7: Конфигурация msk-donskaya-gspolyakov-sw-3 2

Конфигурация msk-donskaya-gspolyakov-sw-4 1

```
msk-donskaya-gspolyakov-sw-4(config)#interface vlan2
msk-donskaya-gspolyakov-sw-4(config-if)#no sh
msk-donskaya-gspolyakov-sw-4(config-if)#no shutdown
msk-donskaya-gspolyakov-sw-4(config-if)#ip ad
msk-donskaya-gspolyakov-sw-4(config-if)#ip address 10.128.1.5 255.255.255.0
msk-donskaya-gspolyakov-sw-4(config-if)#exit
msk-donskaya-gspolyakov-sw-4(config)#ip de
msk-donskaya-gspolyakov-sw-4(config)#ip default-gateway 10.128.1.1
msk-donskaya-gspolyakov-sw-4(config)#line vty 0 4
msk-donskaya-gspolyakov-sw-4(config-line)#pas
msk-donskaya-gspolyakov-sw-4(config-line)#password cisco
msk-donskaya-gspolyakov-sw-4(config-line)#login
msk-donskaya-gspolyakov-sw-4(config-line)#exit
msk-donskaya-gspolyakov-sw-4(config)#line console 0
msk-donskaya-gspolyakov-sw-4(config-line)#password cisco
msk-donskaya-gspolyakov-sw-4(config-line)#login
msk-donskaya-gspolyakov-sw-4(config-line)#exit
msk-donskaya-gspolyakov-sw-4(config)#ena
msk-donskaya-gspolyakov-sw-4(config)#enable se
msk-donskaya-gspolyakov-sw-4(config)#enable secret cisco
msk-donskaya-gspolyakov-sw-4(config)#pas
msk-donskaya-gspolyakov-sw-4(config)#ser
msk-donskaya-gspolyakov-sw-4(config)#service pas
msk-donskaya-gspolyakov-sw-4(config)#service password-encryption
msk-donskaya-gspolyakov-sw-4(config)#user
msk-donskaya-gspolyakov-sw-4(config)#username admin pr
msk-donskaya-gspolyakov-sw-4(config)#username admin privilege 1 secret cisco
msk-donskaya-gspolyakov-sw-4(config)#ip doma
msk-donskaya-gspolyakov-sw-4(config)#ip domai
msk-donskaya-gspolyakov-sw-4(config)#ip domain-n
msk-donskaya-gspolyakov-sw-4(config)#ip do
msk-donskaya-gspolyakov-sw-4(config)#ip dom
msk-donskaya-gspolyakov-sw-4(config)#ip doma
msk-donskaya-gspolyakov-sw-4(config)#ip domain-name donskeya.rudn.edu
msk-donskaya-gspolyakov-sw-4(config)#cry
msk-donskaya-gspolyakov-sw-4(config)#crypto k
msk-donskaya-gspolyakov-sw-4(config)#crypto key g
msk-donskaya-gspolyakov-sw-4(config)#crypto key generate r
msk-donskaya-gspolyakov-sw-4(config)#crypto key generate rsa
```

Конфигурация msk-donskaya-gspolyakov-sw-4 2

```
The name for the keys will be: msk-donskaya-gspolyakov-sw-4.donskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-donskaya-gspolyakov-sw-4(config)#line vty 0 4
*Mar 1 0:27:39.605: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:27:39.605: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-donskaya-gspolyakov-sw-4(config-line)#tran
msk-donskaya-gspolyakov-sw-4(config-line)#transport in
msk-donskaya-gspolyakov-sw-4(config-line)#transport input ss
msk-donskaya-gspolyakov-sw-4(config-line)#transport input ssh
msk-donskaya-gspolyakov-sw-4(config-line)#exit
msk-donskaya-gspolyakov-sw-4(config)#exit
msk-donskaya-gspolyakov-sw-4#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-gspolyakov-sw-4#wr
```

Рис. 9: Конфигурация msk-donskaya-gspolyakov-sw-4 2

Конфигурация msk-pavlovskaya-gspolyakov-sw-1 1

```
msk-pavlovskaya-gspolyakov-sw-1(config)#interface vlan2
msk-pavlovskaya-gspolyakov-sw-1(config-if)#no sh
msk-pavlovskaya-gspolyakov-sw-1(config-if)#no shutdown
msk-pavlovskaya-gspolyakov-sw-1(config-if)#ip add
msk-pavlovskaya-gspolyakov-sw-1(config-if)#ip address 10.128.1.6 255.255.255.0
msk-pavlovskaya-gspolyakov-sw-1(config-if)#exit
msk-pavlovskaya-gspolyakov-sw-1(config)#ip def
msk-pavlovskaya-gspolyakov-sw-1(config)#ip default-gateway 10.128.1.1
msk-pavlovskaya-gspolyakov-sw-1(config)#line vty 0 4
msk-pavlovskaya-gspolyakov-sw-1(config-line)#pass
msk-pavlovskaya-gspolyakov-sw-1(config-line)#password cisco
msk-pavlovskaya-gspolyakov-sw-1(config-line)#login
msk-pavlovskaya-gspolyakov-sw-1(config-line)#exit
msk-pavlovskaya-gspolyakov-sw-1(config)#line console 0
msk-pavlovskaya-gspolyakov-sw-1(config-line)#pass
msk-pavlovskaya-gspolyakov-sw-1(config-line)#password cisco
msk-pavlovskaya-gspolyakov-sw-1(config-line)#login
msk-pavlovskaya-gspolyakov-sw-1(config-line)#exit
msk-pavlovskaya-gspolyakov-sw-1(config)#enable se
msk-pavlovskaya-gspolyakov-sw-1(config)#enable secret cisco
msk-pavlovskaya-gspolyakov-sw-1(config)#ser
msk-pavlovskaya-gspolyakov-sw-1(config)#service pas
msk-pavlovskaya-gspolyakov-sw-1(config)#service password-encryption
msk-pavlovskaya-gspolyakov-sw-1(config)#use
msk-pavlovskaya-gspolyakov-sw-1(config)#username admin pr
msk-pavlovskaya-gspolyakov-sw-1(config)#username admin privilege 1 secret cisco
msk-pavlovskaya-gspolyakov-sw-1(config)#ip domain
msk-pavlovskaya-gspolyakov-sw-1(config)#ip domain-
msk-pavlovskaya-gspolyakov-sw-1(config)#ip domain-name pavlovskaya.rudn.edu
msk-pavlovskaya-gspolyakov-sw-1(config)#cr
msk-pavlovskaya-gspolyakov-sw-1(config)#crypto k
msk-pavlovskaya-gspolyakov-sw-1(config)#crypto key g
msk-pavlovskaya-gspolyakov-sw-1(config)#crypto key generate r
msk-pavlovskaya-gspolyakov-sw-1(config)#crypto key generate rsa
The name for the keys will be: msk-pavlovskaya-gspolyakov-sw-1.pavlovskaya.rudn.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

msk-pavlovskaya-gspolyakov-sw-1(config)#line vty 0 4
*Mar 1 0:30:48.258: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:30:48.258: %SSH-5-ENABLED: SSH 1.5 has been enabled
msk-pavlovskaya-gspolyakov-sw-1(config-line)#tr
msk-pavlovskaya-gspolyakov-sw-1(config-line)#transport i
msk-pavlovskaya-gspolyakov-sw-1(config-line)#transport input s
msk-pavlovskaya-gspolyakov-sw-1(config-line)#transport input ssh
```

Конфигурация msk-pavlovskaya-gspolyakov-sw-1 2

```
msk-pavlovskaya-gspolyakov-sw-1(config-line)#transport input ssh
msk-pavlovskaya-gspolyakov-sw-1(config-line)#exit
msk-pavlovskaya-gspolyakov-sw-1(config)#exit
msk-pavlovskaya-gspolyakov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-pavlovskaya-gspolyakov-sw-1#wri
msk-pavlovskaya-gspolyakov-sw-1#write me
msk-pavlovskaya-gspolyakov-sw-1#write memory
Building configuration...
[OK]
msk-pavlovskaya-gspolyakov-sw-1#
```

Рис. 11: Конфигурация msk-pavlovskaya-gspolyakov-sw-1 2